

Technical Requirements Document (TRD)

DICE by Sanyog — Mobile Application

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1. Technology Stack

1.1 Mobile App

LAYER	TECHNOLOGY	VERSION
Framework	React Native (Expo managed)	SDK 51
Language	TypeScript	5.x
State Management	Zustand	4.x
Navigation	React Navigation	v6
HTTP Client	React Query + Axios	5.x / 1.x
Local Storage	expo-secure-store	13.x
Animations	React Native Animated, LayoutAnimation	Built-in
UI Components	Custom + Ionicons + LinearGradient	—
Push Notifications	expo-notifications	0.28.x
Camera / Scan	expo-camera, expo-barcode-scanner	—
Biometrics	expo-local-authentication	—
File Handling	expo-document-picker, expo-file-system	—

1.2 Backend (Planned)

LAYER	TECHNOLOGY
Runtime	Node.js 20 LTS
Framework	Express.js / NestJS
Database	PostgreSQL 15 (primary), Redis (cache/sessions)
ORM	Prisma
Auth	JWT + Refresh Tokens
File Storage	AWS S3 / Cloudflare R2
Push	Firebase Cloud Messaging (FCM) + APNs
Search	Elasticsearch (for regulations/news)
AI	OpenAI GPT-4 API
Hosting	AWS ECS / Railway

2. Architecture

2.1 App Architecture

```
App.tsx
├─ ThemeProvider (context)
├─ QueryClientProvider
├─ SafeAreaProvider
├─ RootNavigator
│   ├── SplashScreen (native + JS)
│   ├── AuthNavigator (Onboarding → Login → OTP → UserType)
│   └─ DrawerNavigator
│       ├── MainNavigator (Bottom Tabs)
│       │   ├── HomeStackNavigator
│       │   ├── CertificationsStackNavigator
│       │   ├── ApplicationsStackNavigator
│       │   ├── InsightsStackNavigator
│       │   └─ ProfileStackNavigator
│       └─ Additional Stacks (Documents, Payments, Shipment, etc.)
```

2.2 State Management

Zustand Stores:

└─ useAuthStore	– user, token, refreshToken, onboarding state
└─ useThemeStore	– mode: 'light' 'dark' 'system'
└─ useNotificationStore	– notifications[], unreadCount
└─ useBookmarkStore	– bookmarkedIds[]

2.3 Theme System

- Two color palettes: `DarkColors`, `LightColors`
- `ThemeContext` resolves active colors based on mode + system scheme
- `useColorScheme()` drives system detection
- Persisted to `expo-secure-store` under key `scs_theme`
- `UIInterfaceStyle: Automatic` in `Info.plist` for native dark/light

3. Navigation Structure

RootNavigator (Stack)

└─ SplashScreen
└─ OnboardingScreen
└─ LoginScreen
└─ OTPScreen
└─ UserTypeScreen
└─ DrawerNavigator
└─ HomeStack: Home
└─ CertificationsStack: Certifications → CertDetail → CBComparison → CertCenter → CertDetails
└─ ApplicationsStack: Applications → AppDetail → NewApp → TechnicalReview → ...
└─ InsightsStack: Insights → InsightDetail → AIFeed → Circulars → ...
└─ ProfileStack: Profile → EditProfile → Security → Notifications → Language → ...
└─ DocumentsStack
└─ PaymentsStack
└─ ShipmentStack
└─ CommunicationStack
└─ AIStack

4. API Design (Backend)

4.1 Authentication

```
POST /auth/send-otp      { phone }
POST /auth/verify-otp    { phone, otp } → { token, refreshToken, user }
POST /auth/refresh       { refreshToken } → { token }
POST /auth/logout
```

4.2 User / Profile

```
GET /user/me
PUT /user/me             { name, phone, companyName, gstNumber, ... }
PUT /user/onboarding     { businessRole, industries, targetMarkets, ... }
DELETE /user/me
```

4.3 Certifications

```
GET /certifications      ?status=active&type=BIS&page=1
GET /certifications/:id
POST /certifications     { type, productName, standard, ... }
PUT /certifications/:id
DELETE /certifications/:id
GET /certifications/:id/timeline
GET /certifications/expiring ?daysAhead=30
```

4.4 Applications

```
GET /applications        ?status=pending&page=1
GET /applications/:id
POST /applications       { certType, product, labId, ... }
PUT /applications/:id/status
GET /applications/:id/timeline
GET /applications/:id/documents
POST /applications/:id/documents    multipart/form-data
DELETE /applications/:id/documents/:docId
GET /applications/:id/notes
POST /applications/:id/notes        { text }
GET /applications/:id/queries
POST /applications/:id/queries/:qId/respond { response }
```

4.5 Technical Review

```
GET    /applications/:id/technical-review
GET    /technical-reviews/:trId
GET    /technical-reviews/:trId/findings
GET    /technical-reviews/:trId/queries
POST   /technical-reviews/:trId/queries/:qId/respond { response }
PUT    /technical-reviews/:trId                { status }
```

4.6 Documents

```
GET    /documents                                ?appId=&category=
POST   /documents/upload                      multipart/form-data
GET    /documents/:id/download
DELETE /documents/:id
```

4.7 Insights / News

```
GET    /insights                                ?category=BIS&page=1
GET    /insights/:id
POST   /insights/:id/bookmark
DELETE /insights/:id/bookmark
GET    /insights/bookmarks
```

4.8 Notifications

```
GET    /notifications                          ?page=1&unread=true
PUT    /notifications/:id/read
PUT    /notifications/read-all
POST   /notifications/register-token          { token, platform }
```

4.9 Payments

```
GET    /payments                                ?appId=
GET    /payments/:id
POST   /payments/initiate                      { appId, amount, method }
GET    /payments/:id/status
```

5. Data Models

5.1 User

```
{
  id: string (UUID)
  phone: string (unique)
  email?: string
  name: string
  companyName: string
  role: 'admin' | 'manager' | 'viewer'
  gstNumber?: string
  address?: string
  city?: string
  state?: string
  pincode?: string
  subscription: 'free' | 'pro' | 'enterprise'
  businessRole?: string
  industries: string[]
  targetMarkets: string[]
  isVerified: boolean
  createdAt: DateTime
  updatedAt: DateTime
}
```

5.2 Application

```
{
  id: string (UUID)
  appNo: string (auto-generated: SCS-YYYY-XXXX)
  userId: string
  certType: string
  product: string
  standard: string
  status: 'draft' | 'submitted' | 'under_review' | 'lab_testing' |
    'technical_review' | 'approved' | 'rejected' | 'completed'
  progress: number (0-100)
  labId?: string
  assignedTo?: string
  submittedDate: DateTime
  estimatedCompletion?: DateTime
  amount: number
  paidAmount: number
  createdAt: DateTime
  updatedAt: DateTime
}
```

5.3 Certificate

```
{
  id: string
  userId: string
  applicationId?: string
  certNo: string
  type: 'BIS' | 'EPR' | 'WPC' | 'FSSAI' | 'ISO' | 'CE'
  standard: string
  product: string
  issuedDate: DateTime
  expiryDate: DateTime
  issuingAuthority: string
  manufacturer: string
  status: 'active' | 'inactive' | 'expired' | 'revoked'
  createdAt: DateTime
}
```

5.4 TechnicalReview

```
{
  id: string
  trNo: string
  applicationId: string
  status: 'pending' | 'in_progress' | 'completed' | 'failed'
  reviewDate?: DateTime
  reviewedBy: string
  assignedOfficer: string
  labReportNo: string
  labName: string
  summary?: string
  findings: Finding[]
  queries: Query[]
  createdAt: DateTime
}

Finding {
  id: string
  clause: string
  result: 'Pass' | 'Fail' | 'Pending'
  remarks: string
  status: 'pass' | 'fail' | 'pending'
}

Query {
  id: string
  question: string
  raisedBy: string
  date: DateTime
  status: 'open' | 'resolved'
  response?: string
  respondedAt?: DateTime
}
```


6. Security Requirements

REQUIREMENT	IMPLEMENTATION
Auth tokens	JWT (15 min expiry) + Refresh Token (30 days)
Token storage	<code>expo-secure-store</code> (hardware-backed on iOS/Android)
Biometric lock	<code>expo-local-authentication</code> (Face ID / Fingerprint)
API transport	HTTPS only (TLS 1.3)
File uploads	Signed S3 URLs (15 min TTL)
Input validation	Server-side: Zod / class-validator
Rate limiting	100 req/min per IP, 1000 req/min per auth user
OTP	6-digit, 5 min TTL, max 3 attempts

7. Performance Requirements

METRIC	TARGET
App cold start	< 2 seconds
Screen transition	< 300ms
API response (P95)	< 500ms
Image load (cached)	< 100ms
Bundle size	< 30 MB
Memory usage	< 200 MB
List scroll FPS	60 FPS

8. Error Handling

- All API errors return `{ error: string, code: string, statusCode: number }`
- Network errors → offline banner + retry button
- Token expiry → silent refresh, fallback to login
- Unhandled JS errors → `LogBox` in dev, Sentry in production
- Crash reports → Sentry with user context

9. Environment Configuration

```
EXPO_PUBLIC_API_URL=https://api.dicebysanyog.com/v1
EXPO_PUBLIC_SENTRY_DSN=...
EXPO_PUBLIC_GOOGLE_MAPS_KEY=...
EXPO_PUBLIC_OPENAI_KEY=...    # only used server-side
```